HomeWork-3 Report

1. **Crawling [Note down steps implemented for each of the below]**

    a. Any changes in the seed URLs provided?
        - I had to add "https://www.ada.gov/" because a provided link was not allowed by the robots.txt

    b. URL Canonicalization –
        - We basically used url_normalize library. We removed everything behind "#" separately as well.

    c. Frontier Management
        - Used queue.PriorityQueue to add URLs with scores. Then, I rebalanced them every 100 tasks or every 2^i tasks. For rebalancing, I compared with the inlink list to edit the score and make a new queue. I created a separate set() as well so that it is efficient to see if a URL already exists.

    d. Politeness Policy
        - To ensure politeness, I made a dictionary of domains, and I noted down the time it was requested. So, I made sure each task checked if there was a 1-second gap between now and the listed time for each domain.

    e. Document Processing
        - I did link extraction and data extraction separately. For links, I found all "a" tags and only appended to the frontier if the lang="en" (if it exists). I also had to remove certain URLs that were not useful (ex. wikipedia.org/w/index.php was mostly duplicates)
        - Then, for data extraction, I took the "p" tags and removed unnecessary white space and punctuation.

2. **Vertical Search (FOR CS6200 only)**
    a. **Add a Screenshot of your Vertical Search UI**
    b. **Explain briefly how you implemented it.**

3. **Extra Credits Done [Note down what was done for each extra credit]**